

Name:

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Jasmine Torres

TRANSPORT PHENOMENA I, CHE 311

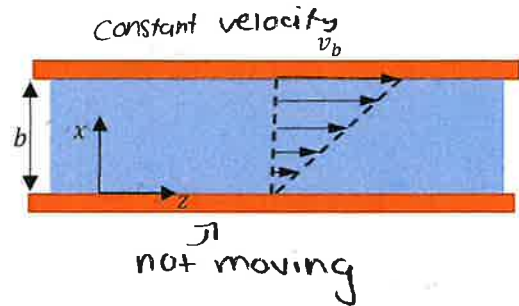
QUIZ 1

CLOSED BOOK, TIME: 15 MINUTES

Imagine a fluid with a constant density ρ between two large parallel plates, separated by a distance b . The top plate is moving with a constant velocity v_b , while the bottom plate is not moving.

The steady-state velocity profile within the fluid is given as

$$v_z = \frac{v_b}{b}x, \quad v_x = v_y = 0$$



- 1- Express the mass flux $\rho\langle v \rangle$ within the fluid.
- 2- Express the convective momentum flux $\rho\langle v \rangle\langle v \rangle$ within the fluid.
- 3- Express the Gradient of velocity $|\nabla v|$ within the fluid.
- 4- Calculate the mass flow rate between the two plates, assuming the width of the plates in y -direction is W .

All answers must be in terms of parameters ρ , v_b , b , and W . You may present your answers for parts 1 through 3 in the matrix format.

$$1) \begin{bmatrix} v_x & v_y \\ v_{bx} & b_x \end{bmatrix} = \begin{bmatrix} 0 \\ v_z \end{bmatrix} \quad 10$$

$\rho\langle v \rangle$

$\rho\langle v \rangle\langle v \rangle$

$|\nabla v|$

Mass flow = W

Name:

Diana Rodriguez

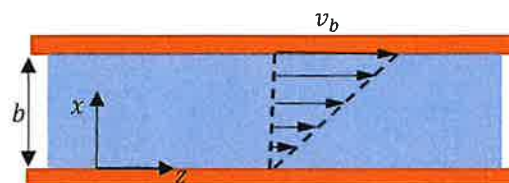
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25 1) $\rho \langle v_x, v_y, v_z \rangle = \rho \langle 0, 0, \frac{v_b}{b} x \rangle$
(dot product) $= \rho \left(\frac{v_b}{b} x \right)$ scalar

10 2) $= \rho \langle v \rangle \langle v \rangle$ vector
 $= \rho \begin{bmatrix} v_x & 0 & 0 \\ 0 & v_y & 0 \\ 0 & 0 & v_z \end{bmatrix} \begin{bmatrix} v_x & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & v_z \end{bmatrix}$

25 3) $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \left(0, 0, \frac{v_b}{b} x \right)$

$\frac{\partial}{\partial x} 0, \frac{\partial}{\partial x} 0, \frac{\partial}{\partial x} \frac{v_b}{b} x = \begin{bmatrix} 0 & 0 & \frac{v_b}{b} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \frac{v_b}{b}$

$\frac{\partial}{\partial y} 0, \frac{\partial}{\partial y} 0, \frac{\partial}{\partial y} \frac{v_b}{b} x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\frac{\partial}{\partial z} 0, \frac{\partial}{\partial z} 0, \frac{\partial}{\partial z} \frac{v_b}{b} x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

0 4)

Name: Patrick Flores

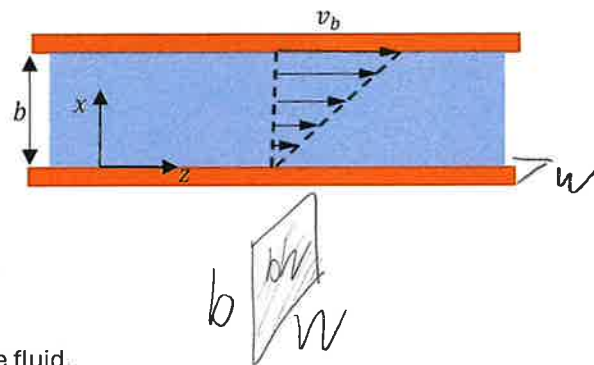
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co-vector

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$$\rho \langle v \rangle$$

$$\langle v \rangle = v_z = \frac{v_b}{b} x$$

$$\rho \langle v \rangle = \rho \cdot \frac{v_b}{b} x$$

$$1.) \rho \begin{bmatrix} 0 & 0 & \frac{v_b}{b} x \end{bmatrix}$$

25

$$2.) \rho \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \left(\frac{v_b}{b}\right)^2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & \frac{v_b}{b} x \\ \frac{v_b}{b} x & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & \frac{v_b}{b} x \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \left(\frac{v_b}{b}\right)^2 x^2 \end{bmatrix}$$

25

$$|\nabla v| = \begin{bmatrix} \frac{dv_x}{dx} & \frac{dv_x}{dy} \\ \frac{dv_y}{dx} & \frac{dv_y}{dy} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{v_b}{b} & 0 & 0 \end{bmatrix}$$

25

$$3.) \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{v_b}{b} & 0 & 0 \end{bmatrix}$$

$$\rho \frac{v_b}{b} x \cdot W b = \rho v_b x W$$

$$4.) \rho \frac{v_b^2}{2} \frac{W b^2}{2}$$

$$\int_0^b \int_0^W \rho \frac{v_b}{b} x \, dx \, dy = \rho v_b \frac{W b^2}{2}$$